

Arctic Geopolitics in a Nutshell? Critical Raw Materials and the Interests of China and the European Union

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Introduction

In this article, we examine a potential avenue of future contestation in the Arctic – namely, the development of critical raw materials (CRMs) – with a particular focus on two actors with a growing interest in exploiting those materials in the Arctic region: China and the European Union (EU). CRMs increasingly play an essential role in the geopolitics of the global energy transition. Today's mining and processing landscape is geographically concentrated around a selected group of countries, such as Australia (for lithium), Chile (copper and lithium), China (graphite and rare earth elements, REEs), the Democratic Republic of Congo (cobalt), Indonesia (nickel) and South Africa (platinum and iridium). And while abundant reserves exist globally, mining and processing capabilities are (still) limited.¹ Moreover, current supply chains are increasingly exposed to several geopolitical risks, such as external shocks, resource nationalism, export restrictions, mineral cartels, political instability and social unrest, as well as market manipulation.²

In comes the Arctic region – a broad geographic area encompassing eight states that is rich in critical minerals, such as lithium, cobalt, nickel and REEs. However, mining in the Arctic poses significant risks, particularly to the region's fragile ecosystem, and would require significant investments, often also involving public-private partnerships. And yet, as demand for critical minerals grows, the Arctic's role in the global supply chain will likely become more prominent, particularly from a Chinese and EU perspective. The Russian invasion of Ukraine in February 2022 marked a watershed in relations between the West and the Russian Federation, including in the Arctic region. Once hailed as an exceptional space of regional governance, cooperation and peaceful co-existence between the Arctic states, that notion lost meaning over the past two – almost three – years. Once used as a case of 'complex interdependence', where (the Arctic) states were able to preserve existing regional cooperation even in cases of international crises, the Arctic of today is increasingly affected by the strategic interactions between the world's superpowers elsewhere. This has both direct and indirect effects on Arctic cooperation – from affecting existing practices of security cooperation to economic cooperation in the (Russian) Arctic through the policy of sanctions, and this may also have an impact on future mining projects in the region.

To discuss the related developments, we will *first* provide an overview of the geopolitics of the global energy transition, looking at the role of CRMs and their importance for a technological-intense industry and economy. *Second*, we will examine how geostrategic incentives and mineral demand drive the involvement of both China and the EU in the Arctic. We briefly look at China's investments in mining operations and its strategic interests in the

region and then analyse the EU's approach to the Arctic, focusing on its demand for Arctic resources and its efforts to secure and diversify access to CRMs. *Finally*, we will analyse what these developments mean for the Arctic region. In doing so, this article will provide some reflections on the implications that the 'geopoliticisation' of CRMs in the Arctic has for the future of the region.

The Geopolitics of the Energy Transition: The Case of Critical Raw Materials

CRMs – elements such as REEs, lithium, cobalt, tungsten and certain other metals – play an instrumental role in today's modern and increasingly changing economies, as they are crucial for supporting technological advancements, economic resilience and environmental sustainability. Today, CRMs are needed for a broad variety of wind energy installations, batteries, catalysts, fuel cells, semiconductors, photovoltaics, fertilisers, magnets, new power grids and modern technologies, including medical applications. As such, these materials are considered essential for the envisaged global energy transition, and related demand will only increase in the decades ahead. A 2020 *World Bank Report*, for example, found that the production of minerals would need to increase by 500 per cent by 2050 to meet the growing demand for clean energy technologies.³ Moreover, some materials are important for defence technologies and national security systems, making the access to and availability of CRMs a key matter for the geoeconomics of the 21st century. The classification of these materials as 'critical' generally considers their economic significance, supply risks and the availability of substitutes. Criticality is determined by how essential a resource is to a region's or state's economy and its vulnerability to supply disruptions due to geopolitical, environmental or market factors. In simple terms, CRMs are 'securitised' raw materials,⁴ indicating how indispensable a material is for various industries and the potential risks associated with its supply disruption.

On a global scale, China is the largest supplier of several CRMs. Today, the country accounts for almost 100 per cent of production of heavy REEs (HREEs), 85 per cent of light REEs (LREEs) and 86 per cent of tungsten.⁵ China's unique position as the largest global player in CRMs is the result of the international division of labour, its effective use of market interventions and the country's industrial policy with long-term investments in specialisations. Even if REE production is diversified outside China, the country will remain in a monopoly position due to its role in the processing and development of REE technologies.⁶ For example, China almost completely dominates the supply chain of rare earth magnets,⁷ which are required for wind generators and electric vehicles.⁸ In addition to its own resources, China also invests heavily in global CRM production and development. For example, in South America, China – like the European states and the United States – invests in mining operations for lithium, copper and other critical minerals.⁹ In Africa, China has built strategic infrastructure, such as roads and railways, to facilitate the transport of minerals such as cobalt and copper, ensuring a steady supply chain back to China for processing.¹⁰ In addition to China, other states are also important global suppliers of specific materials. Russia and South Africa are the largest global

suppliers of platinum group metals, Australia of lithium, the United States of beryllium and helium, and Brazil of niobium.¹¹

On a regional scale, the Arctic already plays an important role in the present supply chain of CRMs and further holds significant potential for future CRM extraction as it contains deposits of 31 out of 34 CRMs.¹² The region is notably rich in REEs such as neodymium,¹³ in addition to cobalt, nickel and platinum group metals. While REEs are relatively abundant in the Earth's crust, minable concentrations are scarce, with North America boasting substantial economically recoverable reserves, including 1.8 million tonnes in the United States, 0.83 million tonnes in Canada, with a significant portion of it in its Arctic regions, and 1.5 million tonnes in Greenland.¹⁴ For Russia, the Arctic accounts for 100 per cent of its REE extraction and over 60 per cent of its deposits, 100 per cent of nickel extraction and 75 per cent of its deposits, 99 per cent of platinoid extraction and 95 per cent of its deposits, 85 per cent of cobalt extraction and 47 per cent of its deposits, 79 per cent of phosphate extraction and 46 per cent of its deposits and 37 per cent of copper production.¹⁵ The European Arctic, particularly the Fennoscandian Shield, which includes Finland, Sweden and Norway, contributes significantly to the EU's domestic production of critical minerals like iron, gold, silver, zinc, cobalt, graphite, lead, mica, nickel, phosphate rock, titanium and copper, and has further potential as a source of these minerals.¹⁶

China: A Key Supplier with Limited Regional Presence

China's Arctic engagement can be traced back to the signing of the Svalbard Treaty in 1925, but it was in the 1990s that active engagement began, then through scientific expeditions. Science has since been a common thread in its Arctic engagement. In 2004, the Polar Research Institute of China opened a research station in the town of Ny-Ålesund on Svalbard, Norway. After China became a 'modern well-off society', a goal it set for itself and reached around 2007, its international activities increased as well. In 2007, China expressed a desire to become an observer on the Arctic Council,¹⁷ a status it was awarded six years later in 2013. At that time, China was welcomed by the European Arctic countries, who reasoned that it was better to include non-Arctic countries rather than to exclude them and risk the formation of alternative clubs. Since 2013, however, China has had a more uphill battle with the Western Arctic countries. In 2018, China published its Arctic policy as a white paper, where it stated that in addition to understanding and protecting the Arctic, China also wanted to participate in Arctic governance, as well as in developing resources and infrastructure.¹⁸ This included minerals, hydrocarbons and shipping routes. China underscored that resource development and environmental protection were both necessary and not mutually exclusive, advocating for sustainable extractive practices in the Arctic.¹⁹ China's Arctic policy was met with scepticism, as the geopolitical landscape shifted around this time, with more pronounced Sino-US rivalry. This became an evident reality in 2019 when then US State Secretary Mike Pompeo gave a thunderous speech at the Arctic Council Kiruna meeting, criticising China and Russia.²⁰ Western-Chinese relations only soured after Russia's invasion of Ukraine in 2022. As the seven other Arctic states – Canada, the Kingdom of Denmark, Finland, Iceland, Norway,

Sweden and the United States – have turned their backs on Russia, for China, the Russian Arctic is where it currently has the most leeway to participate in developing the Arctic. The largest Arctic development projects to date with Chinese ownership are Yamal LNG and Arctic LNG 2, both projects for producing liquefied natural gas located on the Russian Yamal peninsula. China has not changed its Arctic policy, but given the geopolitical circumstances, its regional opportunities for mineral extraction seem limited only to the Russian Arctic.

Regarding CRMs, China's interest in the Arctic is driven by a combination of geostrategic incentives and mineral demand, with official strategic incentives significantly influencing the investment decisions of Chinese companies in the region.²¹ China's assessments of raw materials' strategic nature and criticality are not only different from the West, but also heterogeneous internally and include several levels of criticality.²² At the same time, China's prioritisation has been aimed at securing the country's future dominance in CRMs, such as by tightly controlling the mining, processing and export of REEs.²³ In Greenland, for example, China's interest in mineral resources was welcomed by some but made others wary of potential environmental damage and a loss of control over resources, also raising national security concerns.²⁴ ²⁵ Chinese companies bid for investment in potential mining projects, including a Chinese REE mining company, Shenghe Resources Holding Co Ltd., acquiring a 12.5 per cent²⁶ stake in Greenland Minerals and Energy Ltd., which was planning the Kuannersuit (Kvanefjeld) REE project.²⁷ Although this decision was likely motivated by the domestic distribution of REE quotas in China, the Kingdom of Denmark regarded Chinese investment in Greenland as a potential security threat and incommensurate with efforts to build supply chains independent from China.²⁸ As a result, the Greenlandic government halted the project in 2021.²⁹ In addition to REE projects, Chinese companies have also been involved in lead and zinc mines like Citronenfjord in Greenland and Red Dog Mine in Alaska.³⁰ Due to de-risking strategies, China's ability to participate in mining projects in the Nordic and North American Arctic is (currently) very limited.³¹ In 2022, three Chinese companies, Sinomine Rare Metals Resources Co Ltd., Chengze Lithium International Ltd. and Zangge Mining Investment Co Ltd., were ordered to divest from Canadian lithium mining. Similarly, a new Swedish law mandated foreign direct investment³² screening for 'exploration, extraction, enrichment or sale of critical raw materials or of metals or minerals critical to Swedish supply'.³³ Similarly, a Norwegian government white paper mentioned that 'minerals can be critical' and 'it can therefore be desirable to prevent [that] unwanted foreign actors from gaining access to mineral resources in Norway, on land or on the seabed.'³⁴ Several Western governments, including Canada, Finland, Norway and the United States, created the Minerals Security Partnership, seeking to 'catalyze public and private investment in responsible critical minerals supply chains globally' and diversify from China.³⁵

China's policy response to the Western de-risking strategies is still being formulated.³⁶ Some Chinese scholars see China's position in the REE industry as a 'card' that China can play in its trade competition with the United States.³⁷ In response to the EU's Critical Raw Materials Act (see below), China implemented export controls in 2023 on CRMs, such as gallium,

germanium and certain graphite products.³⁸ In addition, there is growing secrecy around China's mineral resource plans, such as the National Plan for Mineral Resources, which restrict the ability of Western countries to plan de-risking from China.³⁹

Today, China's strategic engagement with Arctic minerals is driven by both structural and strategic reasons. China is committed to the energy transition, with its '30.60 dual goals' of peaking carbon emissions by 2030 and reaching carbon neutrality by 2060. China is the world leader in installed capacity for renewable energy sources,⁴⁰ and is also world-leading in electric vehicles and lithium battery production.⁴¹ These efforts require a steady supply of many minerals and metals. As a result of China's industrial policy, it is the most important actor in the supply chains of many CRMs, including REEs. Therefore, Chinese interest and involvement in any mining project is natural thanks to their expertise and comparative advantages. Furthermore, domestic politics, such as which companies have access to domestic quotas, play a role in the investment decisions of Chinese companies. At the same time, it is in China's interest to maintain control over supply chains. This means that in projects such as those in the Arctic, China has an automatic interest. Since China controls supply chains downstream, such as refining and technology,⁴² additional upstream supply from the Arctic might not be that detrimental to China's position in these supply chains. Furthermore, there is a high cost for investments in the Arctic due to the challenging environmental and legal landscape.⁴³ Moreover, despite current dominance, China may be concerned that supplies could be disrupted by political unrest or pressure from rivals, such as the United States. The battle for critical minerals is 'a new front for strategic competition among global powers', according to China's National Intelligence Agency.⁴⁴

The European Union: A Geopolitical Power with a Growing Demand for CRMs

Over the past 20 years, the EU has also proven to be an active participant in Arctic affairs and has developed a comprehensive approach to the complex Arctic social, economic, political and environmental landscape of the 21st century.⁴⁵ Today, the EU is 'in the Arctic (...) and has strategic and day-to-day interests'.⁴⁶ First of all, the EU has a tangible presence in the region in terms of geography, legal competence and contributions to Arctic science. It also exercises influence due to policies and regulations, both determining access to its single market as well as shaping its environmental footprint in the region. For decades, the EU has participated in different regional forums, such as the Arctic Council and the Barents Euro-Arctic Council. Three EU Member States are Arctic states: the Kingdom of Denmark (on behalf of Greenland and the Faroe Islands, which are themselves outside of the Union), Finland and Sweden. The northernmost regions of Finland and Sweden are subject to the EU's legislation and policies and benefit from EU funding programmes. This influence extends further to Iceland and Norway (with the exception of Svalbard) through the European Economic Area (EEA) Agreement, which applies the majority of the EU's legal norms to these two Arctic states. EU programmes on research and innovation, as well as regional development and cooperation formats (e.g., the Northern Dimension), further extend to the Barents region and North Atlantic basin. While Greenland is not part of the EU, the Union also maintains close relations with this self-governing territory, cooperating in areas such as education, health and fisheries,

and opened an office in Greenland's capital, Nuuk, in March 2024.⁴⁷ Additionally, the EU cooperates on Arctic issues with Canada and the United States, and until 2022, also with regions in northwest Russia. As a major global economy, the EU influences the Arctic via its shared responsibility for climate change through pollution reaching the Arctic from Europe, as well as owing to the EU's demand for Arctic resources. Additionally, the EU influences the development of international norms that are relevant to the Arctic. For instance, EU competencies regarding maritime transport, environmental protection and fisheries have made the Union an important actor in international negotiations and party to agreements and organisations on Arctic maritime navigation, marine biodiversity or the future governance of the Central Arctic Ocean.

From a CRM perspective, the Arctic increasingly plays an important role for the EU (and its Member States) in securing and diversifying access to raw materials. Already in 2008, the European Commission warned that the EU was 'highly dependent on imports of strategically important raw materials'⁴⁸ and started to identify CRMs needed for the EU from 2011 onwards. This list has been updated every three years,⁴⁹ with the latest report published in 2023.⁵⁰ In a 2020 communication, the Commission eventually stated that the 'access to resources is a strategic security question for Europe's ambition to deliver the Green Deal', as these minerals would be crucial to both the green transition and the Union's strategic autonomy.^{51 52} Additionally – and in 2023 – the Commission proposed the Union's Critical Raw Materials Act, a framework effort for ensuring the secure and sustainable supply of CRMs by both internal and external measures in order to strengthen various stages of strategic raw material value chains. The objective outlined by the Act is to reach 10 per cent of the EU's annual needs for extraction, 40 per cent for processing and 25 per cent for recycling by 2030. Additionally, no more than 65 per cent of the Union's annual needs for cobalt, lithium, nickel and REEs (defined as strategic raw materials) at any relevant stage of processing should come from a single third country. Today, the EU is dependent on a few or single-source countries for many of the critical minerals, with Chile providing 79 per cent of lithium, the United States 60 per cent of beryllium and China 100 per cent of HREEs, 85 per cent of LREEs and 97 per cent of magnesium.⁵³ Arctic-related EU imports consist of numerous raw materials, including critical minerals: 69 per cent of gold, 51 per cent of nickel and 48 per cent of titanium.⁵⁴ Greenland's mineral deposits include 25 out of the 34 critical and important raw materials needed for the energy transition.⁵⁵ Recently, the EU and Greenland signed a strategic agreement to establish sustainable value chains to explore these resources.⁵⁶ This cooperation will bring investments in education and skills to Greenland and support the clean energy transition, with the expectation of increased (private) European investments in Greenland. The areas of cooperation include the work on environmental, social and governance standards to ensure that Greenland's mineral sector contributes to sustainable and inclusive economic growth with local and domestic value creation.^{57 58}

Today, the EU is at the forefront of the circular economy and has already increased its use of secondary raw materials. However, while part of the future demand can be met by recovery

and recycling, there are increasing market pressures towards new exploration and primary extraction. Arctic deposits (and, more broadly, deposits in Arctic states) of CRMs would diversify the EU supply. The aim is to ensure that imported resources are extracted as responsibly as possible and do not run counter to local sustainability and livelihoods, although the EU has only limited legal competence and influence with regard to Arctic-regional mineral extraction. Generally, the EU can play different roles with regard to shaping Arctic mineral extraction by being a limited regulator for various aspects of mining activities within the EU and partly in the EEA, supporting the European security of supply and its environmental and social sustainability, encouraging certain types of investments, supporting research and development, as well as by encouraging networking among pan-European industries and authorities.⁵⁹ The EU's raw materials policy is based on several levels of criticality. The term 'critical raw materials' came with the aforementioned 2008 raw materials initiative. In the Critical Raw Materials Act, a definition of strategic raw materials was also included. Strategic raw materials are those materials that are relevant 'for the green and digital transition as well as defence and space applications,' selected and based on demand forecasts and production difficulty. CRMs are defined as any strategic raw material with a high risk of supply disruption. The reason for this is that not all raw materials, for example, nickel and copper, meet the threshold for criticality, which requires a high risk of supply disruption due to the concentration of sources and a lack of good, affordable substitutes.⁶⁰

Arctic Mineral Extraction: The Divide Becomes More Visible

Today, two concurrent trends are affecting mining in the Arctic. On the one hand, the climate crisis and the needed energy transition require increased CRM extraction. On the other hand, intensified geopolitical hostility threatens existing supply chains and supply security. The Arctic is increasingly becoming a region where the United States, Russia, China and the EU are all seeking to reconfigure the global geography of mining.⁶¹ China seeks to keep and expand its position in the global supply chains. The EU is concerned about becoming too dependent on China for CRMs, which has led to efforts to diversify supply chains and reduce reliance on Chinese imports.⁶² The EU and the United States are also aligning their strategies to foster international cooperation on CRMs.⁶³ Norway, in its comments on the EU's Critical Raw Materials Act, stresses the need to 'ensure Europe's access to a stable and sustainable supply of critical raw materials'.⁶⁴ Additionally, Norway and the United States recently signed a Memorandum of Cooperation on critical minerals, with Norway playing an important role in the supply chain by being the 'number one exporter of cobalt to the U.S. [and] the number three exporter of nickel'.⁶⁵ Similarly, Russia also wants to tap into the Arctic 'as a strategic resource base' and use it 'in order to accelerate the economic growth' of Russia.⁶⁶

While considerable attention is directed at the Chinese presence in the Arctic, such as the case in Greenland, it is in fact Western countries like Australia, Canada and the United Kingdom that dominate the mining sector in Greenland.⁶⁷ Similarly, in other parts of the Arctic, it is still the Arctic states that command the extraction of CRMs. Norway's decision to open for concessions on the continental shelf has sparked vigorous discussions on the environmental sustainability of mining in the Arctic seas. At the same time, only opening for

new mining projects upstream will not be detrimental to Chinese control of production chains downstream. In addition to industrial support, one of the reasons that China received the role it has today in the first place was also because of China's willingness to accept negative environmental externalities associated with mineral production with fewer regulatory costs and delays.⁶⁸ De-risking away from China would imply that at least some share of these environmental effects will be borne in the Arctic. There is significant concern about the environmental impact of increased mining activities in the Arctic. The ecosystem in the Arctic is fragile, and large-scale mining projects are likely to aggravate environmental degradation.⁶⁹

Five of the seven Western Arctic states are part of the Minerals Security Partnership.⁷⁰ They are bolstered by other advanced industrial economies, including Australia, the EU, Japan and the Republic of Korea. As this cooperation develops, it will undoubtedly have an effect on Arctic cooperation as well. As the biggest Arctic country, Russia is both increasingly isolated and reliant on China.⁷¹ At the same time, Russia has plans to expand its domestic production of CRMs, such as copper.⁷² Considering its lack of other options, Russia will have to rely on China for technology, extraction, refining and export markets for its Arctic copper. Furthermore, Russia might also rely on China even for exports of CRMs produced by already existing projects, as is the case with Norilsk Nickel's copper refining facilities.⁷³ If Russian CRMs, previously serving the European markets, are indeed directed to China, the pressure on the West to de-risk will be exacerbated.

Where Do We Go from Here?

This brief analysis has looked into how the Arctic region is increasingly becoming critical for the supply of CRMs needed for the global energy transition and how the geopolitics of mining is driving rivalry between China and the EU. One way forward for Arctic countries to handle this new situation could be to coordinate the relevant regulatory standards across borders. With coordination, Arctic countries can be prepared to better handle potential new mineral extraction projects, regardless of where the funding for such projects originates. Such a regulatory framework also needs to take Indigenous rights into account. Resource projects must prioritise the rights and perspectives of Indigenous communities, which can sometimes be sidelined in larger geopolitical strategies. For local Arctic communities, the need for economic activity can weigh heavily, whereas a national government located further south may have other opinions on the matter. The past decade of mineral projects in Greenland springs to mind as an exemplification of this dilemma.

China and the EU both have an interest in accessing materials in the Arctic region. The relationship between China and the EU is important for both countries. There is significant economic interdependence, but trade relations are complex, with tariffs and trade imbalances. Both invest in each other's economies, but there are concerns in the EU about Chinese investments in strategic sectors, leading to calls for more scrutiny and regulation.⁷⁴ Recent imposed tariffs on electric vehicles from China into the EU will further sour the relationship.⁷⁵ Nevertheless, both collaborate on initiatives and agreements, such as environmental, climate change and energy transition efforts.⁷⁶ It seems unlikely that they will join forces in the Arctic in the near future, however. Competing interests are one key aspect.

China's ambitions in the Arctic often focus on resource extraction, while the EU emphasises sustainability and environmental protection, which could lead to conflicts in priorities. Importantly, despite geopolitical tensions between China and the EU, the environment and climate change are areas where dialogue and cooperation are possible, which could alleviate some distrust between them. Trust is another aspect. Historical geopolitical rivalries and differing governance models may hinder effective collaboration and trust-building between China and the EU. This would very much depend on how the geopolitical situation develops in the coming years. The recent introduction of taxes from the EU on electric vehicles will likely escalate the tension between the two, and China may retaliate by introducing taxes on agricultural goods from the EU. Additionally, potential tensions from the partnerships established seeking to counter China's domination of the supply chain for CRMs and China's strained relationship with the United States will likely impact EU–China relations. The initiatives established for developed countries and resource-rich nations to cooperate on projects and policies could develop into trade wars. For the Arctic, geographical isolation does not hinder the wider international situation from impacting what is happening in the north. Debates and deliberations are needed in local communities, nationally and regionally: How can the region, its environment, people and communities be protected, while at the same time also facilitating the energy transition *and* taking geopolitical challenges into account? There are no simple answers, but dialogue is a good place to start.

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